

## CEDAR CITY, UT, USA

WMO: 724755

Lat: 37.709N

Lon: 113.094W

Elev: 1703

StdP: 82.47

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 93129

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | -16.3      | -13.4 | -19.8                         | 0.8 | -10.3 | -17.4 | 1.0 | -6.5 | 12.3                  | 6.7  | 11.1 | 6.0  | 1.0                   | 60   | 0.585 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.8                   | 34.4            | 15.1 | 33.1 | 14.8 | 31.9 | 14.4 | 17.6                | 27.9 | 16.9 | 27.5 | 16.2 | 27.1 | 4.8                  | 210  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 14.6                            | 12.8 | 19.8 | 13.5 | 11.9 | 20.1 | 12.3 | 11.0 | 20.5 | 55.9          | 27.7 | 53.6 | 27.3 | 51.3 | 27.0 | 22.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 11.6          | 10.3 | 8.7 | -21.3                      | 36.8 | 3.2                | 1.0 | -23.7                                              | 37.5 | -25.5      | 38.1 | -27.3      | 38.6 | -29.7      | 39.3 |    |    |
| (5)               |      |     | -21.7                      | 19.0 | 3.2                | 1.0 | -23.9                                              | 19.7 | -25.8      | 20.3 | -27.5      | 20.8 | -29.8      | 21.5 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 10.5   | -0.7  | 1.3   | 5.5   | 8.6   | 13.8  | 19.9  | 23.7  | 22.5  | 17.6  | 10.4  | 4.1   | -1.1  |  |  |
|                                                                        | DBStd       | 9.54   | 5.17  | 4.66  | 4.24  | 4.16  | 4.30  | 3.83  | 2.23  | 2.10  | 3.52  | 4.28  | 4.98  | 5.20  |  |  |
|                                                                        | HDD10.0     | 1388   | 332   | 243   | 146   | 74    | 16    | 1     | 0     | 0     | 2     | 47    | 182   | 345   |  |  |
|                                                                        | HDD18.3     | 3263   | 590   | 477   | 397   | 292   | 149   | 28    | 0     | 1     | 52    | 246   | 428   | 603   |  |  |
|                                                                        | CDD10.0     | 1575   | 0     | 1     | 8     | 33    | 133   | 298   | 423   | 387   | 229   | 59    | 4     | 0     |  |  |
|                                                                        | CDD18.3     | 409    | 0     | 0     | 0     | 0     | 8     | 76    | 165   | 130   | 30    | 1     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 6236   | 0     | 0     | 2     | 17    | 275   | 1403  | 2192  | 1691  | 600   | 56    | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 2716   | 0     | 0     | 0     | 1     | 65    | 626   | 1095  | 753   | 171   | 6     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 3.2    | 2.7   | 3.0   | 3.5   | 3.9   | 3.8   | 3.9   | 3.3   | 3.3   | 3.2   | 2.9   | 2.8   | 2.7   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 286    | 18    | 24    | 30    | 26    | 20    | 12    | 25    | 33    | 24    | 27    | 25    | 17    |  |  |
|                                                                        | PrecMax     | 430    | 76    | 60    | 75    | 80    | 55    | 42    | 111   | 111   | 117   | 85    | 66    | 44    |  |  |
|                                                                        | PrecMin     | 158    | 1     | 0     | 0     | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     |  |  |
|                                                                        | PrecStd     | 68     | 19    | 16    | 19    | 16    | 14    | 10    | 22    | 28    | 24    | 22    | 17    | 11    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 14.8  | 17.7  | 22.6  | 25.6  | 31.0  | 35.4  | 36.5  | 34.8  | 31.9  | 27.4  | 21.2  | 14.2  |  |  |
|                                                                        |             | MCWB   | 5.4   | 6.8   | 8.6   | 10.4  | 13.4  | 14.3  | 15.9  | 15.5  | 13.9  | 11.7  | 8.2   | 5.8   |  |  |
|                                                                        | 2%          | DB     | 11.5  | 14.7  | 19.7  | 23.0  | 28.5  | 33.5  | 35.0  | 33.4  | 30.1  | 25.0  | 18.5  | 11.7  |  |  |
|                                                                        |             | MCWB   | 4.1   | 5.6   | 7.4   | 9.0   | 11.9  | 13.7  | 15.6  | 15.2  | 13.3  | 10.9  | 7.3   | 4.5   |  |  |
|                                                                        | 5%          | DB     | 9.3   | 11.9  | 17.5  | 20.9  | 26.5  | 32.0  | 33.7  | 32.2  | 28.7  | 22.6  | 16.1  | 9.5   |  |  |
|                                                                        |             | MCWB   | 3.3   | 4.3   | 6.5   | 8.3   | 11.0  | 13.2  | 15.3  | 15.0  | 13.1  | 9.8   | 6.3   | 3.4   |  |  |
|                                                                        | 10%         | DB     | 7.1   | 9.4   | 15.2  | 18.7  | 24.3  | 30.5  | 32.3  | 30.9  | 27.2  | 20.5  | 13.4  | 7.2   |  |  |
|                                                                        |             | MCWB   | 2.3   | 3.3   | 5.5   | 7.4   | 10.1  | 12.6  | 14.9  | 14.7  | 12.9  | 8.9   | 5.2   | 2.2   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 6.6   | 7.7   | 9.4   | 11.2  | 14.0  | 15.8  | 18.6  | 18.4  | 17.4  | 13.7  | 9.5   | 6.8   |  |  |
|                                                                        |             | MCDB   | 12.3  | 15.5  | 20.5  | 23.5  | 28.8  | 31.1  | 28.9  | 28.0  | 25.8  | 21.2  | 17.4  | 12.3  |  |  |
|                                                                        | 2%          | WB     | 5.0   | 6.1   | 8.0   | 9.9   | 12.6  | 14.9  | 17.8  | 17.5  | 15.8  | 11.9  | 8.0   | 5.4   |  |  |
|                                                                        |             | MCDB   | 9.7   | 13.2  | 18.3  | 21.0  | 26.0  | 30.6  | 28.3  | 28.0  | 25.3  | 22.4  | 16.6  | 10.0  |  |  |
|                                                                        | 5%          | WB     | 3.9   | 5.0   | 6.9   | 8.8   | 11.6  | 14.0  | 17.1  | 16.8  | 14.9  | 10.7  | 6.8   | 4.0   |  |  |
|                                                                        |             | MCDB   | 8.2   | 10.7  | 16.3  | 19.4  | 24.4  | 29.3  | 27.8  | 27.5  | 24.0  | 20.5  | 14.9  | 8.4   |  |  |
|                                                                        | 10%         | WB     | 2.7   | 3.7   | 5.9   | 7.8   | 10.7  | 13.2  | 16.5  | 16.2  | 13.9  | 9.6   | 5.6   | 2.6   |  |  |
|                                                                        |             | MCDB   | 6.7   | 8.9   | 14.3  | 17.9  | 23.0  | 28.1  | 27.4  | 26.9  | 23.3  | 18.8  | 13.0  | 6.8   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 12.8  | 13.0  | 15.1  | 15.4  | 16.7  | 18.0  | 16.8  | 16.4  | 17.2  | 16.7  | 15.1  | 12.7  |  |  |
|                                                                        |             | MCDBR  | 15.3  | 16.6  | 18.9  | 19.5  | 19.8  | 20.2  | 18.8  | 18.5  | 19.2  | 19.9  | 18.8  | 15.0  |  |  |
|                                                                        |             | MCWBR  | 9.3   | 9.2   | 9.5   | 9.2   | 8.6   | 8.2   | 7.0   | 7.0   | 7.9   | 9.7   | 10.1  | 9.0   |  |  |
|                                                                        |             | MCDBR  | 13.3  | 14.4  | 17.4  | 18.1  | 18.3  | 18.4  | 15.4  | 15.1  | 15.0  | 17.0  | 17.3  | 13.2  |  |  |
|                                                                        | 5% WB       | MCWBR  | 8.4   | 8.4   | 9.1   | 8.8   | 8.0   | 7.7   | 5.6   | 5.7   | 6.5   | 8.4   | 9.7   | 8.2   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.244 | 0.259 | 0.270 | 0.296 | 0.305 | 0.303 | 0.343 | 0.345 | 0.298 | 0.287 | 0.253 | 0.248 |  |  |
|                                                                        | taud        |        | 2.607 | 2.555 | 2.555 | 2.477 | 2.500 | 2.510 | 2.457 | 2.435 | 2.575 | 2.564 | 2.628 | 2.592 |  |  |
|                                                                        | Ebn at Noon |        | 972   | 990   | 1003  | 984   | 974   | 971   | 929   | 922   | 958   | 942   | 953   | 942   |  |  |
|                                                                        | Edh at Noon |        | 70    | 84    | 93    | 107   | 107   | 106   | 111   | 110   | 90    | 82    | 68    | 66    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2.77  | 3.61  | 5.09  | 6.41  | 7.32  | 8.29  | 7.13  | 6.57  | 5.79  | 4.29  | 3.11  | 2.39  |  |  |
|                                                                        | RadStd      |        | 0.24  | 0.34  | 0.33  | 0.36  | 0.55  | 0.51  | 0.41  | 0.40  | 0.31  | 0.35  | 0.18  | 0.19  |  |  |

## Historical Trends

|                             | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only           |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (7 neighbors) |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page